

20 Lecture 20: Quantum Fourier Transform (QFT)¹

20.1 Discrete Fourier Transform²

The discrete Fourier transform is defined as a transformation on N complex numbers $y_0, y_1, \dots, y_{N-1}$ yielding N complex numbers $z_0, z_1, \dots, z_{N-1}$ defined by:³

$$z_k \equiv \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} y_j e^{2\pi i jk/N} \quad (20.1) \quad ^4$$

Note the analogy with the usual Fourier transform of a function $f(x)$:⁵

$$\tilde{f}(k) = \frac{1}{\sqrt{2\pi}} \int dx f(x) e^{ikx} \quad (20.2) \quad ^6$$

where the sum on the discrete index j is replaced by an integral on the continuous “index” x , and y_j and z_k correspond respectively to $f(x)$ and $\tilde{f}(k)$.⁷

20.2 Quantum Fourier Transform⁸

It is a unitary operation QFT defined on basis vectors $|0\rangle, |1\rangle, \dots, |N-1\rangle$ as⁹

$$QFT|j\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i jk/N} |k\rangle \quad (20.3) \quad ^{10}$$

Then the components y_j of a vector $|v\rangle = \sum_{j=0}^{N-1} y_j |j\rangle$ transform under QFT into new components z_k given by the discrete Fourier transform (prove it). We will show in the following that QFT is indeed unitary.¹¹

Using n qubits, the dimension N of the vector space of physical states is 2^n , and the basis vectors $|0\rangle, \dots, |j\rangle, \dots, |2^n - 1\rangle$ are given by the 2^n vectors of the computational basis $|00 \dots 0\rangle, \dots, |j_1 j_2 \dots j_n\rangle, \dots, |11 \dots 1\rangle$, where $j_1, j_2, \dots, j_n$ is the binary expression of the integer j .¹²

20.3 Binary numbers and binary fractions¹³

The binary representation of the integer j is $j_1 j_2 \dots j_n$ where¹⁴

$$j = j_1 2^{n-1} + j_2 2^{n-2} + \dots + j_n 2^0 \quad (20.4) \quad ^{15}$$

Real numbers between 0 and 1 can be represented with *binary fractions*, defined by¹⁶

$$0.j_1 j_2 \dots j_n \equiv \frac{j_1}{2} + \frac{j_2}{2^2} + \dots + \frac{j_n}{2^n} \quad (20.5) \quad ^{17}$$

and related to the binary integers $j_1 j_2 \dots j_n$ by¹⁸

$$0.j_1 j_2 \dots j_n = \frac{1}{2^n} j_1 j_2 \dots j_n \quad (20.6) \quad ^{19}$$

20.4 Product representation of QFT ¹

Theorem: the unitary operator QFT can be written as the product ²

$$QFT|j_1j_2\cdots j_n\rangle = \frac{1}{2^{\frac{n}{2}}}(|0\rangle + e^{2\pi i \cdot 0.j_n}|1\rangle)(|0\rangle + e^{2\pi i \cdot 0.j_{n-1}j_n}|1\rangle) \cdots (|0\rangle + e^{2\pi i \cdot 0.j_1\cdots j_n}|1\rangle) \quad (20.7) \quad 3$$

Proof: ⁴

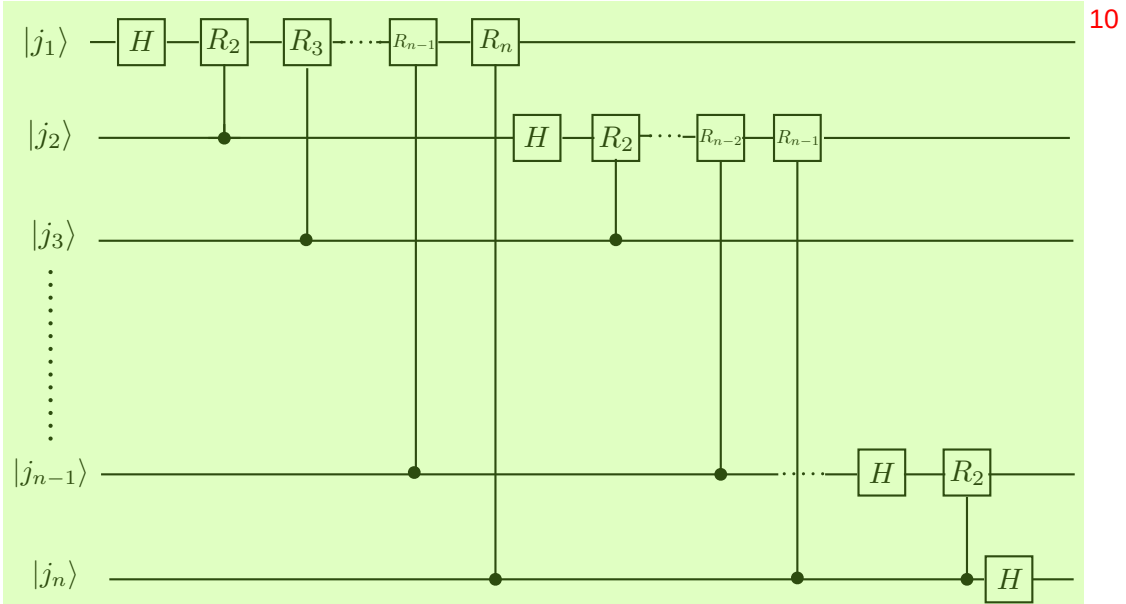
$$\begin{aligned} QFT|j\rangle &= \frac{1}{2^{\frac{n}{2}}} \sum_{k=0}^{2^n-1} e^{2\pi i \cdot jk/2^n} |k\rangle = \frac{1}{2^{\frac{n}{2}}} \sum_{k_1=0}^1 \cdots \sum_{k_n=0}^1 e^{2\pi i \cdot j/2^n (k_1 2^{n-1} + \cdots + k_n 2^0)} |k_1 \cdots k_n\rangle \quad 5 \\ &= \frac{1}{2^{\frac{n}{2}}} \sum_{k_1=0}^1 e^{2\pi i \cdot jk_1 2^{-1}} |k_1\rangle \sum_{k_2=0}^1 e^{2\pi i \cdot jk_2 2^{-2}} |k_2\rangle \cdots \sum_{k_n=0}^1 e^{2\pi i \cdot jk_n 2^{-n}} |k_n\rangle \\ &= \frac{1}{2^{\frac{n}{2}}} (|0\rangle + e^{2\pi i \cdot j/2} |1\rangle)(|0\rangle + e^{2\pi i \cdot j/2^2} |1\rangle) \cdots (|0\rangle + e^{2\pi i \cdot j/2^n} |1\rangle) \quad (20.8) \end{aligned}$$

and observing that ⁶

$$e^{2\pi i \cdot j/2} = e^{2\pi i \cdot 0.j_n}, \quad e^{2\pi i \cdot j/2^2} = e^{2\pi i \cdot 0.j_{n-1}j_n}, \dots, e^{2\pi i \cdot j/2^n} = e^{2\pi i \cdot 0.j_1\cdots j_n} \quad (20.9) \quad 7$$

the theorem follows. $\square$ ⁸

20.5 Efficient circuit for QFT ⁹



where ¹

$$R_k \equiv \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{pmatrix} \quad (20.10) \quad ^2$$

For example $R_1 = Z$, $R_2 = S$, $R_3 = T$, and in general $R_k^2 = R_{k-1}$. ³

The above circuit implements the QFT on the n -qubit input state $|j\rangle = |j_1\rangle \cdots |j_n\rangle$. ⁴

Proof: after the first H gate, the input state becomes (prove it): ⁵

$$\frac{1}{2^{\frac{1}{2}}}(|0\rangle + e^{2\pi i \cdot 0 \cdot j_1}|1\rangle) |j_2\rangle \cdots |j_n\rangle \quad (20.11) \quad ^6$$

Applying next the controlled- R_2 gate produces the state (prove it): ⁷

$$\frac{1}{2^{\frac{1}{2}}}(|0\rangle + e^{2\pi i \cdot 0 \cdot j_1 j_2}|1\rangle) |j_2\rangle \cdots |j_n\rangle \quad (20.12) \quad ^8$$

After the application of all controlled- R gates on the first qubit the state has become ⁹

$$\frac{1}{2^{\frac{1}{2}}}(|0\rangle + e^{2\pi i \cdot 0 \cdot j_1 j_2 \cdots j_n}|1\rangle) |j_2\rangle \cdots |j_n\rangle \quad (20.13) \quad ^{10}$$

Applying the same procedure to the second qubit leads to: ¹¹

$$\frac{1}{2}(|0\rangle + e^{2\pi i \cdot 0 \cdot j_1 j_2 \cdots j_n}|1\rangle) (|0\rangle + e^{2\pi i \cdot 0 \cdot j_2 \cdots j_n}|1\rangle) |j_3\rangle \cdots |j_n\rangle \quad (20.14) \quad ^{12}$$

and repeating it on all the other input qubits yields the final state that coincides with the output of $QFT|j_1 j_2 \cdots j_n\rangle$ in (20.7), once we invert the order of the output qubits with CROSSOVER gates. $\square$ ¹³

The circuit proves that QFT is unitary, since all gates are unitary. The circuit uses $n + (n-1) + \cdots + 1 = n(n+1)/2 = \Theta(n^2)$ gates. In contrast, the best classical algorithms for the discrete Fourier transform use $\Theta(n2^n)$ gates, so that QFT has an exponential advantage on its classical counterpart. ¹⁴